

Show query box

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0001 seconds.)

```
-- ===== -- L06: Constraints & Triggers -- COMP4018 Final Project - F1 Database -- Professor César Bolaños --
-- ===== USE f1_db;
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0011 seconds.)

```
-- ----- -- SECTION 1: CHECK CONSTRAINTS -- Validate data at the attribute/tuple level --
-- ----- -- CHECK: points must be >= 0 in RESULT ALTER TABLE RESULT ADD CONSTRAINT chk_result_points CHECK
(points >= 0);
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
-- CHECK: positionOrder must be > 0 in RESULT ALTER TABLE RESULT ADD CONSTRAINT chk_result_position CHECK (positionOrder > 0);
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0009 seconds.)

```
-- CHECK: grid position must be >= 0 (0 = pit lane start) ALTER TABLE RESULT ADD CONSTRAINT chk_result_grid CHECK (grid >= 0);
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
-- CHECK: laps must be >= 0 ALTER TABLE RESULT ADD CONSTRAINT chk_result_laps CHECK (laps >= 0);
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
-- CHECK: points >= 0 in DRIVER_STANDING ALTER TABLE DRIVER_STANDING ADD CONSTRAINT chk_dstand_points CHECK (points >= 0);
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
-- CHECK: wins >= 0 in DRIVER_STANDING ALTER TABLE DRIVER_STANDING ADD CONSTRAINT chk_dstand_wins CHECK (wins >= 0);
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
-- CHECK: points >= 0 in CONSTRUCTOR_STANDING ALTER TABLE CONSTRUCTOR_STANDING ADD CONSTRAINT chk_cstand_points CHECK (points >= 0);
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
-- CHECK: wins >= 0 in CONSTRUCTOR_STANDING ALTER TABLE CONSTRUCTOR_STANDING ADD CONSTRAINT chk_cstand_wins CHECK (wins >= 0);
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
-- CHECK: lap number must be > 0 in LAP_TIME ALTER TABLE LAP_TIME ADD CONSTRAINT chk_lap_number CHECK (lap > 0);
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

```
-- CHECK: stop number must be > 0 in PIT_STOP ALTER TABLE PIT_STOP ADD CONSTRAINT chk_stop_number CHECK (stop > 0);
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

```
-- CHECK: driver code must be exactly 3 characters ALTER TABLE DRIVER ADD CONSTRAINT chk_driver_code CHECK (code IS NULL OR CHAR_LENGTH(code) = 3);
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

```
-- ----- -- SECTION 2: AUDIT LOG TABLE -- Used by AFTER trigger to record all inserts --
-- ----- CREATE TABLE IF NOT EXISTS RESULT_AUDIT_LOG ( logID INT AUTO_INCREMENT PRIMARY KEY, action_type
VARCHAR(10) NOT NULL, resultID INT, raceID INT, driverID INT, constructorID INT, points FLOAT, positionOrder INT, logged_at TIMESTAMP DEFAULT
CURRENT_TIMESTAMP ) ENGINE=InnoDB;
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0010 seconds.)

```
-- SECTION 3: BEFORE INSERT TRIGGER -- Validates data BEFORE it enters the database -- Prevents
negative points from being inserted -- DROP TRIGGER IF EXISTS before_result_insert;
```

Note: #1360 Trigger does not exist

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0124 seconds.)

```
CREATE TRIGGER before_result_insert BEFORE INSERT ON RESULT FOR EACH ROW BEGIN -- Prevent negative points IF NEW.points < 0 THEN SIGNAL SQLSTATE
'45000' SET MESSAGE_TEXT = 'INTEGRITY VIOLATION: points cannot be negative in RESULT'; END IF; -- Prevent position order of 0 or less IF
NEW.positionOrder <= 0 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'INTEGRITY VIOLATION: positionOrder must be greater than 0'; END IF; --
Prevent negative laps IF NEW.laps < 0 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'INTEGRITY VIOLATION: laps cannot be negative'; END IF; END;
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0001 seconds.)

```
-- SECTION 4: AFTER INSERT TRIGGER -- Runs AFTER a successful insert into RESULT -- Logs the insert
into RESULT_AUDIT_LOG -- DROP TRIGGER IF EXISTS after_result_insert;
```

Note: #1360 Trigger does not exist

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0106 seconds.)

```
CREATE TRIGGER after_result_insert AFTER INSERT ON RESULT FOR EACH ROW BEGIN INSERT INTO RESULT_AUDIT_LOG ( action_type, resultID, raceID, driverID,
constructorID, points, positionOrder ) VALUES ( 'INSERT', NEW.resultID, NEW.raceID, NEW.driverID, NEW.constructorID, NEW.points, NEW.positionOrder
); END;
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0001 seconds.)

```
-- SECTION 5: BEFORE UPDATE TRIGGER -- Validates data BEFORE an update is applied --
DROP TRIGGER IF EXISTS before_result_update;
```

Note: #1360 Trigger does not exist

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0095 seconds.)

```
CREATE TRIGGER before_result_update BEFORE UPDATE ON RESULT FOR EACH ROW BEGIN IF NEW.points < 0 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT =
'INTEGRITY VIOLATION: points cannot be negative on UPDATE'; END IF; IF NEW.positionOrder <= 0 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT =
'INTEGRITY VIOLATION: positionOrder must be > 0 on UPDATE'; END IF; END;
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
-- SECTION 6: TRANSACTION EXAMPLES -- BEGIN, COMMIT, ROLLBACK (TCL) --
-- EXAMPLE 1: Successful transaction - insert a new driver -- All or nothing: both PERSON and DRIVER
inserted together START TRANSACTION;
```

1 row inserted. (Query took 0.0003 seconds.)

```
INSERT INTO PERSON (personID, forename, surname, nationality, dob) VALUES (9999, 'Test', 'Driver', 'Puerto Rican', '2000-01-01');
```

1 row inserted. (Query took 0.0002 seconds.)

```
INSERT INTO DRIVER (driverID, driverRef, number, code, url) VALUES (9999, 'test_driver', 99, 'TST', '');
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
COMMIT;
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0000 seconds.)

```
-- Both rows committed successfully -- EXAMPLE 2: Rollback on error - clean up test driver START TRANSACTION;
```

1 row deleted. (Query took 0.0002 seconds.)

```
DELETE FROM DRIVER WHERE driverID = 9999;
```

1 row deleted. (Query took 0.0002 seconds.)

```
DELETE FROM PERSON WHERE personID = 9999;
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0002 seconds.)

```
COMMIT;
```

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Your SQL query has been executed successfully.

```
-- Test driver removed cleanly --
-- SECTION 7: VERIFY TRIGGERS EXIST --
SHOW TRIGGERS FROM f1_db;
```

Trigger	Event	Table	Statement	Timing	Created	sql_mode	Definer	character_set_cler
before_result_insert	INSERT	RESULT	BEGIN -- Prevent negative points IF NEW.poin...	BEFORE	2026-06-08 16:17:54.80	STRICT_TRANS_TABLES,ERROR_FOR_DIVISION_BY_ZERO,NO_...	root@localhost	utf8mb4
after_result_insert	INSERT	RESULT	BEGIN INSERT INTO RESULT_AUDIT_LOG (actio...	AFTER	2026-06-08 16:17:54.84	STRICT_TRANS_TABLES,ERROR_FOR_DIVISION_BY_ZERO,NO_...	root@localhost	utf8mb4
before_result_update	UPDATE	RESULT	BEGIN IF NEW.points < 0 THEN SIGNAL SQLSTA...	BEFORE	2026-06-08 16:17:54.87	STRICT_TRANS_TABLES,ERROR_FOR_DIVISION_BY_ZERO,NO_...	root@localhost	utf8mb4

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

```
--
-- SECTION 8: TEST THE BEFORE INSERT TRIGGER -- This should FAIL with an integrity error --
(comment out after testing) --
-- UNCOMMENT TO TEST (will throw error as expected): -- INSERT INTO
RESULT (resultID, raceID, driverID, constructorID, grid, positionOrder, points, laps, status) -- VALUES (99999, 1, 1, 1, 1, 1, -5, 10, '1'); --
Expected error: INTEGRITY VIOLATION: points cannot be negative in RESULT --
-- SECTION 9: CHECK AUDIT LOG AFTER AN INSERT --
-- Run this after any valid insert into RESULT to see audit trail: --
SELECT * FROM RESULT_AUDIT_LOG ORDER BY logged_at DESC LIMIT 10;;
```